CSCE 670 · DATA MINING & ANALYSIS · FINAL PROJECT

Instacart Market Basket Analysis: Sequential & Associative Pattern Mining

Combining FP-Growth and PrefixSpan to uncover the hidden temporal rhythm of grocery shopping behavior across 3.4 million orders.

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Texas A&M University · Spring 2026

The Grocery Blind Spot

Why today's recommendation engines are fundamentally incomplete

- **Single-Basket Thinking:** Current systems only analyze what's in today's cart.
  FP-Growth finds co-purchases but ignores the order of purchases over time.
- **No Temporal Memory:** They ignore the purchasing timeline of individual users.
  Flour on Monday → eggs next week — no current system captures this.
- **The Cost:** Wasted perishable inventory and millions in missed revenue.
  Our EDA confirmed >99.9% sparsity — most items in <1% of baskets.

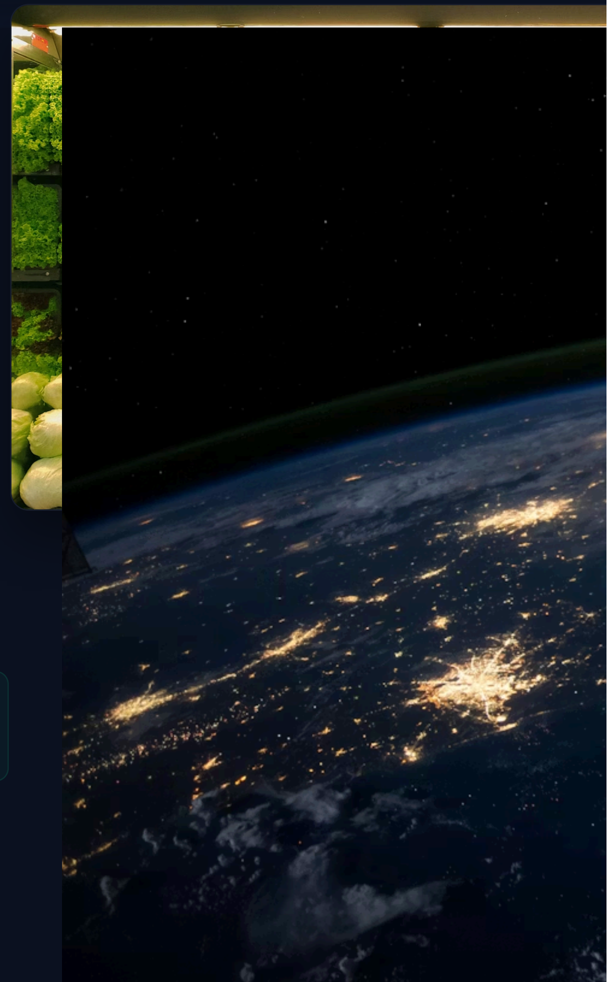
Predicting the Future Cart

Two complementary mining approaches on real Instacart data

3.4M	200K+	50K+
ORDERS	USERS	PRODUCTS

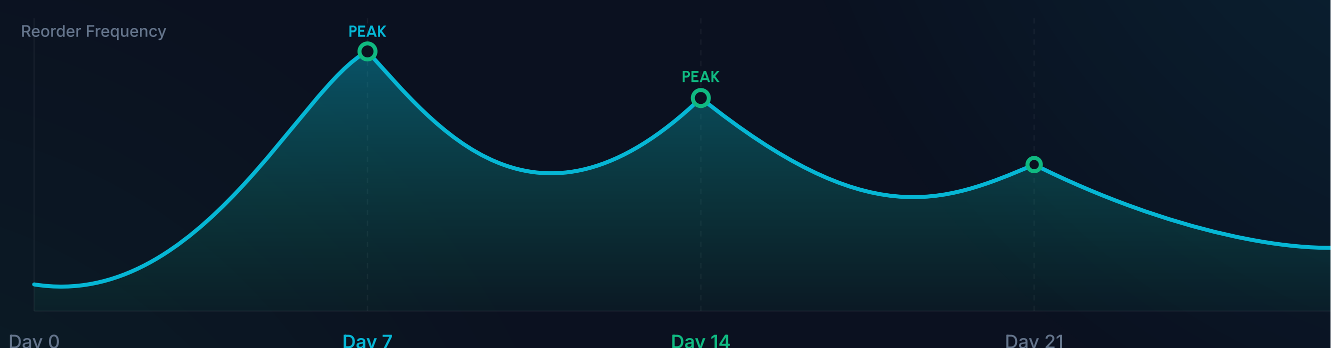
- **Real-World Scale:** 778K+ transactions across 5,000 sampled users.
  Reproducible sampling (seed=42) from the 3.4M order Instacart dataset.
- **Two-Pronged Approach:** FP-Growth for co-occurrence + PrefixSpan for sequences.
  Course technique: "bought together" · External: "bought next".
- **Basket Trajectories:** Chronological sequence of canonicalized basket tuples.
  Proper nested structure preserving order boundaries.

WEEK 1: 🍌 Banana, 🥛 Milk → WEEK 2: 🍌 Banana, 🥚 Eggs → WEEK 3: 🍌 Banana, 🍓 Berries



The 7 & 14-Day Rhythm

EDA on 5,000 sampled users reveals strict, predictable shopping cycles



- Distinct reorder spikes at **7, 14, 21, and 30 day** intervals — strict weekly rhythms.
- Over **99.9% sparsity** — most of 50K products appear in <1% of baskets.
- Early-week vs late-week shoppers sl basket compositions.

What We Found & Where It Goes

Concrete results from 778K+ real transactions

344 **Association Rules** via FP-Growth at 0.5% support across 56K baskets

10K+ **Sequential Patterns** mined by PrefixSpan across basket tra

3+ **Multi-Step Trajectories** — basket-to-basket-to-basket sequences only PrefixSpan can capture

3 **Research Questions** fully answered: thresholds, segments, sequen

The Takeaway: PrefixSpan captures multi-step purchasing trajectories that co-occurrence analysis structurally cannot repres ordered sequence of *what comes after what*.

This trajectory structure enables next-basket prediction and demand forecasting beyond what static basket analysis can provide.

The Opportunity: Pre-stock warehouses using 7-day cycle forecasting, push perfectly timed coupons, build time-aware recommendations.

Transform millions of grocery receipts into a roadmap for predictive behavioral logistics.